

DEMO-201:2026

Illustrative Protective Helmets for Two-Wheeler Riders Specification (synthetic demo document, not an Indian Standard)

Third Revision - ICS 43.14

FOREWORD

This is a synthetic demonstration document created for the BISense prototype. It is not an Indian Standard, it has not been issued by the Bureau of Indian Standards, and all values in it are illustrative only.

This illustrative specification covers protective helmets worn by riders and passengers of two-wheeler motor vehicles. It sets out construction requirements, performance tests for impact absorption and penetration, and marking requirements.

1 SCOPE

1.1 This document specifies the requirements for protective helmets for riders and passengers of two-wheeler motor vehicles.

1.2 This document does not cover helmets for industrial use or for sports such as cycling.

2 REFERENCES

The following documents contain provisions which, through reference in this text, constitute provisions of this document:

DEMO-900 Illustrative Rules for Rounding Off Numerical Values

DEMO-910 Illustrative Headforms for Helmet Testing

3 TERMINOLOGY

For the purpose of this document, the following definitions shall apply.

3.1 Shell - The hard outer part of the helmet that gives it its general shape.

3.2 Protective Padding - The material inside the shell that absorbs impact energy.

3.3 Retention System - The complete assembly, including the chin strap, that holds the helmet in position on the head.

3.4 Visor - A transparent protective screen extending over the eyes and covering part of the face.

4 REQUIREMENTS

4.1 Construction

4.1.1 The helmet shall consist of a shell, protective padding, comfort padding and a retention system.

4.1.2 The shell shall not have any external projection exceeding 5 mm above its outer surface.

4.1.3 The helmet should have a visor for protection against wind and debris.

4.2 Mass

The mass of the complete helmet shall not exceed 1.2 kg.

4.3 Field of Vision

The helmet shall give a horizontal peripheral vision of not less than 105 degrees on each side of the median plane.

4.4 Performance Requirements

The helmet shall meet the performance requirements given in Table 1 when tested as prescribed in clause 8.

Table 1 Performance Requirements for Helmets

SI No.	Test	Requirement	Method of Test
i	Impact absorption, peak acceleration, Max	275 g	Clause 8.2
ii	Penetration	Striker shall not contact the headform	Clause 8.3
iii	Retention system, dynamic elongation, Max	35 mm	Clause 8.4
iv	Rigidity, deformation under load, Max	40 mm	Clause 8.5

4.5 Materials

4.5.1 Materials that come into contact with the skin shall not be known to cause skin irritation.

4.5.2 The shell may be made of thermoplastic or fibre-reinforced composite material.

5 PACKING

5.1 Each helmet shall be packed in a carton that protects it from damage during transport.

6 MARKING

6.1 Each helmet shall be permanently and legibly marked with the following information:

- a) Name of the manufacturer;
- b) Size in centimetres;
- c) Month and year of manufacture; and
- d) The words "Synthetic demo label" for this illustrative document.

6.2 Each helmet shall be accompanied by instructions for use and care.

7 SAMPLING

7.1 For type testing, eight helmets of each size shall be selected at random from a lot.

7.2 The lot shall be declared conforming if all helmets pass all the tests in Table 1.

8 TEST METHODS

8.1 Conditioning - Before testing, helmets shall be conditioned at ambient, high temperature (50 degrees C) and low temperature (minus 20 degrees C) for 4 hours.

8.2 Impact Absorption - The helmet, fitted on a headform, shall be dropped onto a flat anvil and the peak acceleration of the headform shall be recorded.

8.3 Penetration - A pointed striker of mass 3 kg shall be dropped onto the helmet from a height of 1 m.

8.4 Retention System - A mass of 10 kg shall be dropped through a height of 750 mm to load the chin strap.

8.5 Rigidity - The helmet shall be compressed between two parallel plates with a load of 630 N.

AMENDMENT NO. 1 MARCH 2026

(Page 2, clause 4.2) - Substitute "1.2 kg" for "1.5 kg". This illustrative amendment demonstrates how BISense records amendments; it does not describe any real revision.